Business processes and Use cases

V1.1

1. Business Process

1.2 Definition

A **business process** is a structured sequence of activities or tasks performed by people, systems, or both to achieve a specific organizational goal. It describes **how work is performed within a business environment** to produce a product or service.

Business processes focus on **the operational workflow of an organization**, independent of any specific software system.

In system analysis, business processes are often documented using techniques such as:

- **Business Process Model and Notation (BPMN)**
- **Process flow diagrams**
- **Workflow diagrams**

1.3 Characteristics of a Business Process

A business process typically:

- Has a **clear objective**
- Involves **multiple steps or activities**
- May involve **multiple actors or departments**
- Produces a **measurable outcome or output**
- Exists **regardless of whether a software system is used**

1.4 Example 1 – Online Retail Store

Business Process: Order Fulfillment

Steps in the process may include:

1. Customer places an order
2. System verifies payment
3. Warehouse receives order request
4. Items are picked from inventory
5. Package is prepared
6. Courier collects the parcel
7. Customer receives shipment notification

The **entire workflow** from order placement to delivery is the **business process**.

1.5 Example 2 – Hospital Patient Registration

Business Process: Patient Admission

Typical steps:

1. Patient arrives at hospital
2. Reception captures patient details
3. System checks if patient exists in database
4. Patient file is created or retrieved
5. Medical record is updated
6. Patient is assigned to a doctor

This sequence represents the business process, regardless of what software system supports it.

2. Use Case

2.2 Definition

A **use case** describes a **specific interaction between a user (actor) and a system** to achieve a particular goal.

A use case focuses on **how a system behaves** when a user performs an action.

Use cases are typically documented using:

- **Use Case Diagrams (UML)**
- **Use Case Specifications**
- **User stories or functional requirements**

2.3 Characteristics of a Use Case

A use case:

- Represents **one specific function of a system**
- Involves **an actor interacting with the system**
- Produces a **defined system outcome**
- Is usually part of a **larger business process**

2.4 Example 1 – Online Retail Store

Within the **Order Fulfillment business process**, the system may support several use cases.

Use Case: Place Order

Actor: Customer

Steps:

1. Customer logs into the website
2. Customer selects products
3. Customer adds items to cart
4. Customer confirms order
5. System processes payment
6. System generates order number

This interaction between **customer** and **system** is a **use case**.

2.5 Example 2 – Hospital System

Within the **Patient Admission business process**, several use cases exist.

Use Case: Register Patient

Actor: Receptionist

Steps:

1. Receptionist opens registration screen
2. Receptionist captures patient details
3. System validates information
4. System creates patient record
5. System assigns patient ID

This represents a use case supported by the hospital information system.

2.6 Key Differences Between Business Process and Use Case

Aspect	Business Process	Use Case
Level	High-level organizational workflow	System-level interaction
Focus	How the organization performs work	How users interact with the system
Scope	May involve people, systems, and manual tasks	Focused only on system functionality
Actors	Departments, employees, systems	User roles interacting with a system
Representation	Process diagrams, BPMN, workflow diagrams	UML use case diagrams and descriptions
Relationship	Contains multiple use cases	Supports part of a business process

2.7 Summary

- A **business process** describes the **overall workflow of an organization**.
- A **use case** describes **how a user interacts with a system to perform a specific function**.
- **Business processes define the operational environment**, while **use cases define system functionality that supports those operations**.
- Multiple **use cases typically support a single business process**.